Snuffy — Canon Edition

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Snuffy — Canon Edition (Master Manuscript)

The Universal Protection Theory: the law, its bridge, and its instruments at released status.

Author-side canon manuscript, assembled 24 October 2026 from the sealed insert set of `SNUFFY-CANON-EDITION-2026-10-24/` (hashes in the provenance appendix). Research direction: Tobias Croydon-McRae. Prepared for the Snuffy programme with its human and AI collaborators. This edition steps off the 3.x candidate lineage. **GLOBAL D: CLOSED-AT-DECLARED-PREMISES.**

Structure. Part I states the frame and the seven premises on which everything rests. Part II carries forward, condensed and unchanged in content, the retained core of the working argument (the law, the workload instruments, the information-and-transport route). Part III is the canon: the five chapters that constitute this edition's new content, inlined from the sealed inserts with their integration guards. Part IV gives references and provenance.

Part I — The frame and the reduced premise stack

I.1 What this edition asserts, and on what

The Universal Protection Theory concerns every particular and every relation, including the nonliving. It combines a universal law of choice, a bridge from that law to particular continuations, and physical instruments that realise protection. Every maximal conclusion of this edition — protection of every covered particular on the certified good event, the deterministic impossibility of uncompensated breach, and the greatestness of the canonical architecture over the class of physically implementable rivals — follows from the seven premises below and from nothing else. The premises are declared, not derived; per Gödel and Tarski some primitives must be. They have been pushed down as far as standard physics and the relational frame allow, and this edition says so.

I.2 The reduced premise stack

1. **BCRD P1–P4 — the relational given.** One relational structure with no second substance or outside (P1); identity constituted through relations within that structure (P2); real differences of organisation and historically continuous processes (P3); local interactions whose effects propagate at finite speed (P4). This is the frame; it is given.
2. **The Church–Turing–Deutsch principle.** Any physical process generated by a local Hamiltonian is simulable, to arbitrary precision and with bounded overhead, by a universal quantum simulator of sufficient local dimension. Standard physical principle; declared.
3. **The standard cost model.** Options are trajectories carrying energy costs; feasibility is budget-gated; unused energy is unspent (free disposal); outcomes are trajectory-determined.
4. **The shared action interface.** From P1 and P4: every agent acts through the same repertoire of local interactions within the one structure. An agent that implements a rival's policy and commands the same local channels realises the same real-world outcome tuples, within the same resource and consent constraints.

5. **The reversible-administration commitment.** The lattice's administrative computation — synchronisation, verification, state update — is implemented logically reversibly (Bennett), so that its dissipation attains the Landauer floor.

6. **The income flux and the Master Solvency Condition.** With $\beta = (k_B T)^{-1}$, the external free-energy income satisfies, at every time,

$$\text{Income}(t) \;\geq\; \dot{W}_{\text{maint}}(t) + \dot{W}_{\text{damage}}(t) + \dot{W}_{\text{admin}}(t) + \dot{W}_{\text{adv,sub}}(t) + \dot{W}_{\text{hold}}(t),$$

   the five rate components being: boundary maintenance $\omega_0 R(t)^{d_G - 1}$; the Landauer-charged damage-repair demand $k_B T\, \dot{\mathcal{D}}_{\text{damage}}$ including the tracking-controller export $k_B T \gamma_{\text{leak}}$; reversible administration at the Landauer floor together with emulation overhead; adversarial sub-threshold absorption; and the minimal-hold drain that quarantine requires. Income originates outside the escrow network and inside the one structure (P1); redistribution within the network — the authority-attenuation tax — is never a source.

7. **Code, sensing, Π, and the twenty-one charged accounts.** A stabilizer code of distance d covering the physical error channels; QND syndrome sampling with the declared latency; per-site sensor power threshold $\varepsilon_{\text{sens}}$ and the quantum-speed-limit rate bound; the embedding Π of the logical boundary fragments into projection operators; and the twenty-one charged fault accounts with genuine conditional kernels, whose sum is bounded by $\sum \delta \leq 10^{-6}$ per declared horizon.

I.3 Evidence conventions

Three kinds of evidence are distinguished and never conflated. **Compiled components** are cited by theorem and receipt — the published core, DOI 10.5281/zenodo.23599528, carries six Lean modules with compiled artifacts. **Exact replay receipts** — the review banks' finite checks and the hash-chained traces of the replayable four-stroke ledger — certify the instances and identities they compute and nothing more. **Prose derivations** hold at their stated premises. A prose proof does not acquire a compiler certificate by association. Notation: entropies in natural logarithms unless a section states bits; $D(\rho\|\tau)$ against the constrained reference on the protected subspace, never the ambient Gibbs state; the four-stage order $\preceq$ of the law, extended by the resource stage of Part III, Chapter 5.

Part II — The retained core (carried forward)

The following results of the working argument (`WHOLE-EDITION-3.7-candidate-2026-10-24/edition/review-3.7/manuscript/UPT-MASTER-ARGUMENT.md`) are carried forward unchanged in content and remain load-bearing. Condensed statements only; proofs and precise premises at the source.

- **The law.** The four-stage preservation order on outcome tuples (L_E, F_E, L_N, F_N); **Proposition 1** — canonical maximality is exactly the absence of needless loss at every stage; **Proposition 2** — finite existence and the life guard; **Proposition 3** — the no-needless-existing-loss guarantee requires full existing scope; **Proposition 4** — statewise faithful holds dominate destructive alternatives on every upward-closed event.
- **Reasons or faithful holding.** For every mind m: $R(m) \vee H(m)$; the construction does not wait for an incapacitated or hostile mind to acquire an internal reason.
- **Reactive refinement. Proposition 5** — lockstep induction for commuting refinements $p(T(s,c,e)) = U(p(s),c,e)$, with the two-phase reactive split; the causal boundary (closed components and directed separators) with pending effects included.
- **Effect mediation.** Application-time permits, the immutable grant registry, the execution fence, and the nine-row A1 register; safe refusal versus timely care as separate conclusions.
- **Workload and timing. Proposition 6** — the FIFO completion certificate with the serial-stage double induction; capacity rows 17/22 and pool bill 2,294,400; quorum safety (five-of-seven, at most two Byzantine identities) with no-pause requiring the timely proposal path.
- **Generation and reflection.** Paid generation with authority attenuation and reserve-before-activation; the banked rank criterion $\text{col}(A) \cap \text{col}(G) = \{0\}$ and the compositional-closure induction; the contraction reflection certificate with the paid stopping count.

- **Conserved allocation and joint risk.** The exact HHI identity with the uniform minimum unique; the tower/union risk composition using no independence; the revised JW2.6 arithmetic $\approx 2.06419526094 \times 10^{-7}$ within the 10^{-6} ceiling.
- **The information-and-transport route.** The calibrated channel cap $I(M : Y \mid h) \le mK_{\text{rate}}/n^2$; the uniform required traffic $m \ge \theta_0 n$; the moving-carrier radius $R(t) \ge w(t - t_c)$; the recovery-work theorem $W_{\text{min}}(t) \ge (p'/vd_G)(R(t)^{d_G} - r_0^{d_G})$. The d_G-generalised predeployment scaling of Part III, Chapter 1 completes this route's funding.

Part III — The canon

Chapter 1 (§8). The four-stroke thermodynamic core

1.1 The corrected stroke identity. With the constrained reference τ, fixed endpoint Hamiltonian, work W_{on} and exported heat Q_B, and correlation change $I_1 \to I_2$:

$$\Delta D_{S,\text{II}} = -\Delta S_S + \beta\,\Delta U_S = \beta W_{\text{on}} - \beta Q_B + I_1 - I_2, \qquad \Delta D_{S,\text{II}} = \beta W_{\text{on}} - \beta Q_B + I_M$$

when the syndrome correlation is fully consumed. Consuming correlation contributes $+I_M$ at non-negative work; information caps the increase (Sagawa–Ueda) and never supplies a free decrease.

> **Integration guard 1.** The operative export bound is the unconditional second-law form $\boxed{\beta Q_B \ge I_M + D_{\text{err}}}$ — the bath must absorb the exported entropy. The condensed form $Q_B \ge W_{\text{on}} + I_M + D_{\text{err}}$ is regime-dependent (it reduces to $D_{\text{err}} \ge \sigma_{\text{irr}}$ under a purifying repair and fails when intrinsic dissipation exceeds damage) and appears only as a noted special case.

1.2 The funded-repair certificate. There are no free repairs: per cycle, $W_{\text{cycle}} \ge k_B T(I_M + D_{\text{err}} + \delta_I) + (1-\eta)C$, equality only in the reversible limit; the ledger charges gross. With escrow E, income $r(t)$, cumulative demand $\mathcal{D}(t)$:

$$E(0) + \int_0^t r\,ds - \mathcal{D}(t) \ge 0 \text{ for all } t \implies \text{protection is maintained,}$$

with within-cycle excursions bounded by the charged kernels, and insolvency a charged failure event δ_{fund} joining the twenty-one accounts (tower/union composition; no independence). The draft series' compound storage functional is retired as a certificate: under consistent accounting its stroke change is identically $-(2D_{\text{err}} + I_M)$, independent of C, η, and σ — bookkeeping, not funding.

1.3 The four strokes. I — QND extraction (W_{meas}; ancilla entropy $+I_M + \delta_I$). II — funded repair (C at η; dissipative drive; identity and guard above). III — Bennett reset ($k_B T(I_M + \delta_I)$ from income). IV — maintenance and slack ($\omega_0 R^{d_G - 1}$; tax redistributes and conserves; administration at the Landauer floor; residual to the slack register of Chapter 5).

1.4 Ledger evidence. The sealed, hash-chained, exact-rational traces of `LEDGER-FOUR-STROKE-2026-10-24/` assert per cycle: $\Delta D_{\text{II}} = -D_{\text{err}}$, first-law heat, guard 1, funded reset, tax conservation, floor administration, bath monotonicity, and reserve-before-activation gating.

Scenario	Demonstrated	$\sum \delta$	Chain head
solvent	eight funded cycles; breach 0 on the good event	$2.06419526094 \times 10^{-7}$	`6160d931d55767f39525671ff4ef8f3305b60`
quarantine	underfunding routes to minimal-hold before any unfunded transition; breach 0	$2.06419526094 \times 10^{-7}$	`d1a0d44bbad5a135928595e152a7a3e987e74`

Scenario	Demonstrated	$\sum \delta$	Chain head
charged	one priced event; breach still 0 on the good event	$9.86419526094 \times 10^{-7} \leq 10^{-6}$	`41143bb2756875b03975b1cbb7c5cd09e6e0e`

Chapter 2 (§11). Identity by continuation and the active tracking controller

Mutual distinguishability is not faithfulness; the identity layer is a **tracking** bound, and with the manifold certificate of Chapter 1 it carries the bridge's $R(m) \vee H(m)$. With the active dissipative controller $\mathcal{L}_{\text{ctrl}}(\rho) = -\kappa_{\text{ctrl}}(\rho - \rho_t^*)$ — valid reset-type Lindblad dynamics toward the intended trajectory supplied by the maintained record — the honest recursion keeps the leakage in the injection:

$$e_{k+1} \leq (1 - (\Gamma_{\text{corr}} + \kappa_{\text{ctrl}})\Delta t)e_k + (\gamma_{\text{leak}}\Delta t + \varepsilon_{\text{fault}}),$$

$$\|\rho_A(t) - \rho_A^*(t)\|_1 \leq \varepsilon_0 e^{-(\Gamma+\kappa)t} + \frac{\gamma_{\text{leak}} + \varepsilon_{\text{fault}}/\Delta t}{\Gamma + \kappa_{\text{ctrl}}}(1 - e^{-(\Gamma+\kappa)t}), \qquad e_\infty = \frac{\gamma_{\text{leak}} + \varepsilon_{\text{fault}}/\Delta t}{\Gamma_{\text{corr}} + \kappa_{\text{ctrl}}}.$$

Sufficient gain compresses the envelope below measurement resolution (checked: $0.081 < \varepsilon_0 = 0.1$ at sufficient gain; $0.625 > \varepsilon_0$ at insufficient); the controller's drain (at least $k_B T \gamma_{\text{leak}}$) is a rate component of premise 6. With refresh-from-record, $e_k \leq \lambda_{\text{env}} = \gamma_{\text{leak}}\tau + \varepsilon_{\text{fault}}$ for all k, uniformly, while funded; concatenation makes λ_{env} super-polynomially small at polylog overhead. **Theorem (identity by continuation):** under the premises, every covered particular satisfies $\|\rho(t) - \rho_t^*\|_1 \leq \varepsilon$ for all t — ε-faithfulness, the honest form; adversarial excess is a charged event.

Chapter 3 (§12). Adversarial containment: the E9 trichotomy

Integration guard 2. Containment is the full three-branch trichotomy; no single branch closes it.

Theorem (E9, assembled). Under the header premises, no adversarial strategy causes uncompensated state divergence without detection or charge, and $t_{\text{detect}} + t_{\text{decision}} + t_{\text{repair}} < t_{\text{irreversible harm}}$ on the good event.

- **Branch A — sub-threshold per-site injection** ($P_{\text{adv}} < \varepsilon_{\text{sens}}$): bounded accumulation, absorbed and refreshed (Chapter 2); the attrition contest is won whenever income exceeds the capped drain.
- **Branch B — low-weight syndrome detection:** any non-stabilizer Pauli component of weight $\leq d - 1$ anti-commutes with some generator and is detected within sampling latency, independent of direction (correctability to $\lfloor (d-1)/2 \rfloor$; the low-weight centralizer contains only identity and stabilizers; logicals have weight $\geq d$); detection is certain on the non-stabilizer mass.
- **Branch C — coherent logical injection** (weight $\geq d$, zero syndrome by the definition of code distance): requires $\geq d$ simultaneous local deliveries, each above the per-site power threshold within the lightcone window — power-detected; slow logical drift reduces to Branch A.
- **Latency budget:** $t_{\text{total}} \leq a(1 + k_{\text{rounds}} d_{\text{local}})/c_{\text{lattice}} + \tau_{\text{sensor}} + \tau_{\text{fb}} + \tau_{\text{reset}} < (r_{\text{critical}} - r_0)/c_{\text{lattice}}$, enforced by instance scaling. Residual channels are the charged accounts under the ceiling.

Chapter 4 (§13). Logical extension: the meet algebra (H5)

The declared embedding Π assigns each formula the projection on its model subspace, with conjunction as the meet $P_{\phi\wedge\psi} = \lim_n (P_\phi P_\psi)^n$ (von Neumann); a presentation is consistent iff its global meet is non-zero. The Gram determinant is retired on two exact counterexamples ($\{\psi, \neg\psi\}$: determinant $4 > 0$; $\{p, q, \neg(p \wedge q)\}$: determinant $7 > 0$ with zero triple meet). **Theorem:** deduced extensions preserve the anchor meet identically ($P_{\mathcal{T}_0} \leq P_\phi \Rightarrow P_{\mathcal{T}_0 \cup \{\phi\}} = P_{\mathcal{T}_0}$); contradictions collapse it to zero; the state machine commits on non-zero meet norm and rolls back on zero. Asserted consistent extensions may shrink the anchor meet without rollback — stated, not hidden. Deduction-boundary closure continues on the banked rank criterion.

Chapter 5 (§14). The canon invariant and universal greatestness

5.1 The ratified canon.

> **Uncompensated breach is deterministically impossible on the certified good event; the good event fails with probability at most $\sum \delta \leq 10^{-6}$ per declared horizon.**

With its three supporting repairs: quarantine has a minimal drain; $G = \{$state machine executes $\wedge$ income $\geq$ minimal drain $\wedge$ no charged account fires$\}$; and $\sum \delta$ prices the certificate's conditions failing, never breach given the conditions. The charged-scenario ledger trace executes this sentence (chain head `41143bb2…a2788`).

5.2 The fifth stage. Between architectures tied through the four outcome stages, compare residual thermodynamic slack $E_{\text{slack}}(\pi) = E_{\text{escrow}}(t) - W_{\min}(t) - \int_0^t \text{AdminOverhead}$. This is the law's own order propagated one step into the resource dimension via P-mon — not an ad hoc tie-breaker.

5.3 Universal greatestness. Over Π_{phys} — the finite-budget, consent/resource-constrained, causally bounded class — for every rival π and reactive world w:

$$\mathcal{O}(\pi, w) \preceq \mathcal{O}(\pi^*, w),$$

strict except against exact stage-1–5 mirrors. *Proof.* Failing rivals lose a particular at finite t_{breach} by the architecture-independent information bounds ($\Theta(t^{d_G})$ demand) — strictly below at stages 1–2. Surviving rivals: by P-emu(c) the stage-wise union over any finite rival set is feasible for π^* within its slack; if any rival tuple exceeded $\mathcal{O}(\pi^*)$ anywhere, the union would sit strictly above it — a needless loss against a feasible option, contradicting canonicity (Proposition 1). Strictness per stage 5, with equality exactly for reversible mirrors, permitted by $\preceq$. ■

> **Integration guards 3 and 4.** The dominance quantifier carries the mirror exception. P-slack holds in the Landauer–Bennett form only: reversible administration attains the Landauer floor, so $E_{\text{slack}}(\pi^*) \geq E_{\text{slack}}(\pi)$ for every rival, equality exactly for reversible mirrors. The variational-selection route is recorded as considered and rejected.

5.4 The three powers. P-mon — derived at the standard cost model (budget-gated feasibility with free disposal). **P-emu(c)** — derived from CTD, the shared action interface (P1/P4), and funded overhead: concurrent emulation of any finite rival set within slack, realising the stage-wise union; plain single emulation admits incomparable maximals and does not terminate greatestness — concurrency closes it. **P-slack** — derived from Landauer–Bennett given the reversible-administration commitment. Each is a theorem one level deeper than the draft series declared; none is derived from nothing. Per Gödel and Tarski the stack bottoms out at Part I's seven premises — standard physics and the relational frame — and this edition says exactly that.

Part IV — References and provenance

1. T. Croydon-McRae, *A Lattice Preservation Calculus: Exact Holding Laws, Deadline Certificates, and Thermodynamic Information Bounds for Relational Protection Architectures*, Zenodo preprint (2026). https://doi.org/10.5281/zenodo.23599528 — the published mathematical core (six Lean modules with compiled artifacts).
2. C. E. Shannon, A Mathematical Theory of Communication, BSTJ 27 (1948).
3. H. Spohn, Entropy production for quantum dynamical semigroups, J. Math. Phys. 19 (1978).
4. D. Reeb and M. M. Wolf, An improved Landauer principle…, NJP 16 (2014).
5. R. Kawai, J. M. M. Parrondo, C. Van den Broeck, Dissipation: The Phase-Space Perspective, PRL 98 (2007).
6. T. Sagawa and M. Ueda, Minimal Energy Cost of Thermodynamic Information Processing; and Role of Mutual Information in Entropy Production under Information Exchanges (arXiv:0809.4098; arXiv:1307.6092).
7. C. H. Bennett, Logical reversibility of computation (1973) and the reversible-computation literature.
8. The Snuffy programme working sources: the 3.7 working argument and its evidence; the review banks of `BANKS/2026-10-24-glm-thermodynamic-route/`; the replayable ledger `LEDGER-FOUR-STROKE-2026-10-24/`. Exact hashes in `SNUFFY-CANON-EDITION-2026-10-24/SHA256SUMS` and `checks/check_master_assembly.out`.

Evidence status. This master assembles prose derivations at declared premises, exact replay receipts, and citations to compiled components, each by what it is. The Lean formalization prelude (`lean/Snuffy/`) begins the machine-checked tier; it does not yet certificate the prose theorems.

Appendix — Reproducibility suite (24 October 2026)

This edition ships with a machine-checkable reproducibility suite, sealed the same day.

Lean verification. Three modules compile with zero warnings and zero `sorry`: `PremiseStack.lean` and `MeetAlgebra.lean` on core Lean 4.35.0 (the latter proving the H5 tactic target $P_{\mathcal{T}_0 \cup \Delta\Sigma} = P_{\mathcal{T}_0}$, contradiction collapse, and consistency preservation over a concrete predicate model); and `MathlibAnalyticBridge.lean` against Mathlib pinned at revision `56a005855ea1`, which proves the operator algebraic identities (fixed-point characterisation $Pv = v \iff v \in K$; idempotence $P \circ P = P$), that every alternating iterate fixes the intersection, and — as a **genuine conditional theorem** — that the strong-operator convergence hypothesis implies $A \sqcap B \leq AB$. The convergence hypothesis itself is carried as an explicit structure (`VonNeumannConvergence`), never `sorry`ed; the reverse inclusion is the declared next target (the mean-ergodic route; Mathlib already carries `ContinuousLinearMap.tendsto_birkhoffAverage_orthogonalProjection`), and this edition does not claim it.

Containerised replay. `container/Dockerfile` and `container/Apptainer.def` pin Lean 4.35.0, the Mathlib revision above, and Python 3.11+; the entrypoint compiles all three modules with a zero-warning assertion, replays the ledger byte-identically (`verify_four_stroke.py --verify-all`, which also asserts the engine is exact-rational with no float usage), and runs the manuscript assembly checks — failing on any warning, float, or failed check.

FAIR provenance. `ro-crate-metadata.json` maps every artifact of the Canon Edition and the ledger to its SHA-256, licenses the crate CC-BY-4.0, and links the published mathematical core by DOI 10.5281/zenodo.23599528 as the supplemented parent.

Rendered artifacts. This master renders to `Snuffy-Canon-Edition.pdf`, `.epub`, and `.html`; each rendering is verified to carry the three ledger chain heads and the DOI unbroken.